Power Bi Module End Assignment

Bike Stations Analysis Using Power BI

Project Overview and Objective:

- This project analyzes a **bike-sharing station dataset** to evaluate **station availability, operational status, and geographic distribution** across regions, helping stakeholders understand system performance and coverage.

- To monitor **open vs closed station status** and overall network health, To analyze **bike availability rate** for operational efficiency
- To identify **high- and low-coverage regions** using geographic mapping, To support **data-driven decisions** for station expansion and maintenance planning.

Problem Statement:

- Lack of clear visibility into **real-time bike availability across stations** affects user satisfaction.
- High number of stations makes it difficult to **track operational status (open vs closed) efficiently**.
- Uneven **geographical distribution of active stations** may lead to service gaps in certain regions.
- Decision-makers lack **data-backed insights** to prioritize station maintenance and expansion.
- Without performance analysis, it is hard to **optimize bike utilization and reduce downtime**.

Attribute (column /Features) Details:

Column Name	**Data Type**	**Description**
Station_id	Integer / Text	Unique identifier assigned to each bike station.
Address	Text	Physical address of the bike station.

Banking	Boolean (Yes/No)	Indicates whether the station supports card or banking payments.
bonus	Boolean (Yes/No)	Shows if the station is part of a bonus or promotional program.
Status	Text	Operational status of the station (e.g., *Active* or *Inactive*).
contract_name	Text	Name of the city or contract under which the station operates.
bike_stands	Whole Number	Total number of bike stands available at the station.
available_bike_stands	Whole Number	Number of empty stands available for parking bikes.
available_bikes	Whole Number	Number of bikes currently available for use.
last_update	Date/Time	Timestamp of the most recent data update for the station.
city	Text	City where the bike station is located.
latitude	Decimal Number	Geographic latitude coordinate of the station.
longitude	Decimal Number	Geographic longitude coordinate of the station.

Tools and Technologies:

- **Power BI**: Data cleaning, transformation, Data modelling, DAX calculations, visualization, and interactive dashboard creation.

Date Pre-Processing (Power Query)

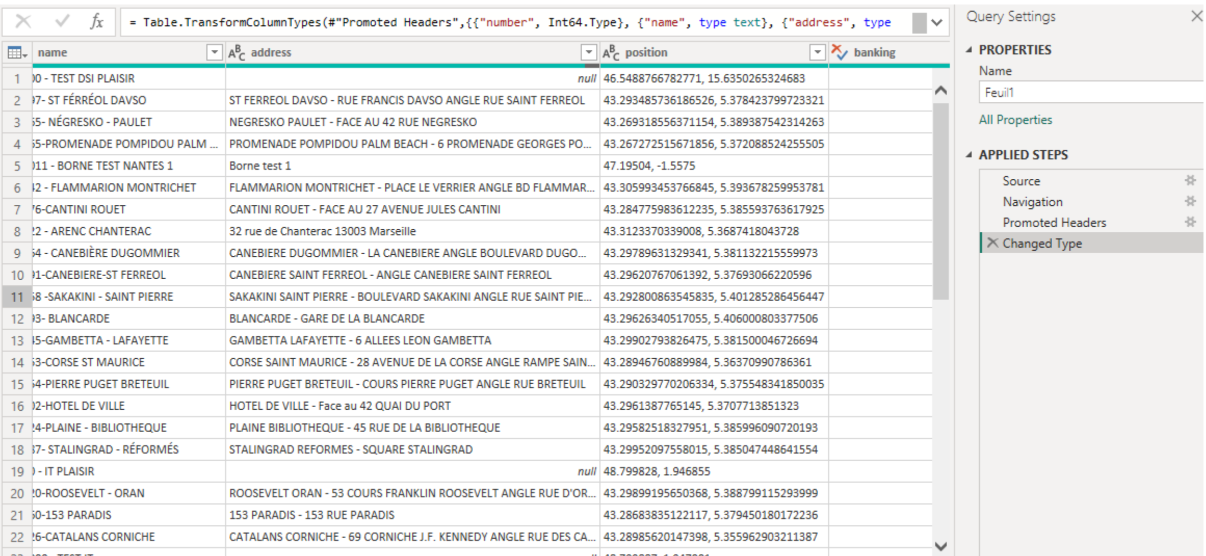
Task Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.

Data Modelling and DAX(Power BI)

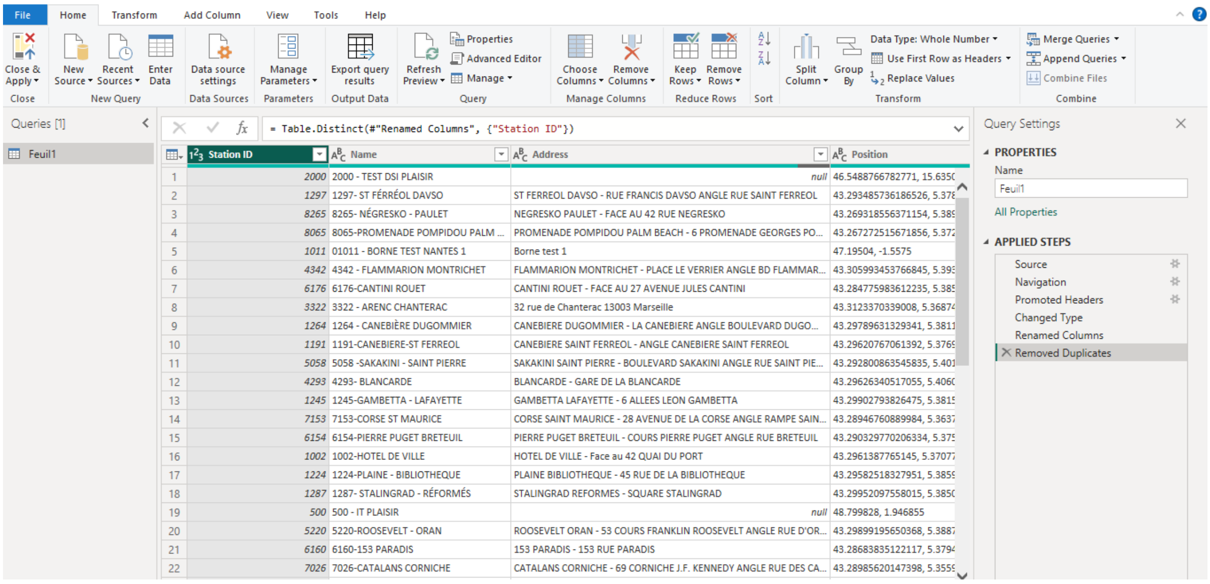
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as Total Stations, Active Stations, Total Bike Stands, Total Available Bikes, Total Available Bike Stands, Bike Utilization %, bike availability Rate %

Transform to Power query

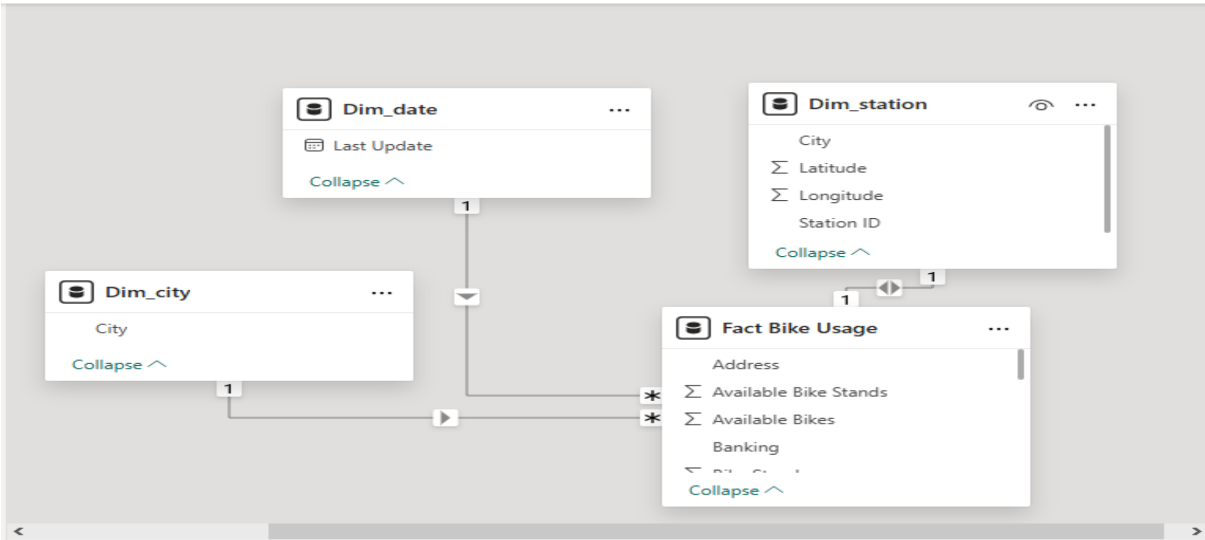


	name	address	position	banking
1	)0 - TEST DSI PLAISIR	null	46.5488766782771, 15.6350265324683	
2	)7- ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	43.293485736186526, 5.378423799723321	
3	55- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	43.269318556371154, 5.389387542314263	
4	55-PROMENADE POMPIDOU PALM ...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	43.267272515671856, 5.372088524255505	
5	)11 - BORNE TEST NANTES 1	Borne test 1	47.19504, -1.5575	
6	12 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...	43.305993453766845, 5.393678259953781	
7	76-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI	43.284775983612235, 5.385593763617925	
8	22 - ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille	43.3123370339008, 5.3687418043728	
9	54 - CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...	43.29789631329341, 5.381132215559973	
10	)1-CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL	43.29620767061392, 5.37693066220596	
11	58 -SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...	43.292800863545835, 5.401285286456447	
12	)3- BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE	43.29626340517055, 5.406000803377506	
13	15-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA	43.29902793826475, 5.381500046726694	
14	53-CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAIN...	43.28946760889984, 5.36370990786361	
15	54-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL	43.290329770206334, 5.375548341850035	
16	)2-HOTEL DE VILLE	HOTEL DE VILLE - Face au 42 QUAI DU PORT	43.2961387765145, 5.3707713851323	
17	24-PLAINE - BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE - 45 RUE DE LA BIBLIOTHEQUE	43.29582518327951, 5.385996090720193	
18	37- STALINGRAD - RÉFORMÉS	STALINGRAD REFORMES - SQUARE STALINGRAD	43.29952097558015, 5.385047448641554	
19	) - IT PLAISIR	null	48.799828, 1.946855	
20	20-ROOSEVELT - ORAN	ROOSEVELT ORAN - 53 COURS FRANKLIN ROOSEVELT ANGLE RUE D'OR...	43.29899195650368, 5.388799115293999	
21	50-153 PARADIS	153 PARADIS - 153 RUE PARADIS	43.28683835122117, 5.379450180172236	
22	26-CATALANS CORNICHE	CATALANS CORNICHE - 69 CORNICHE J.F. KENNEDY ANGLE RUE DES CA...	43.28985620147398, 5.355962903211387	

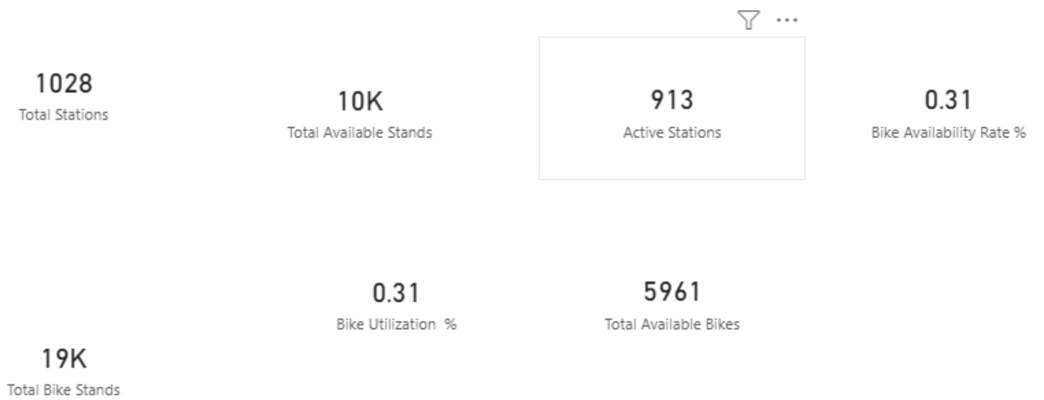
Removed Duplicates



Data Modelling

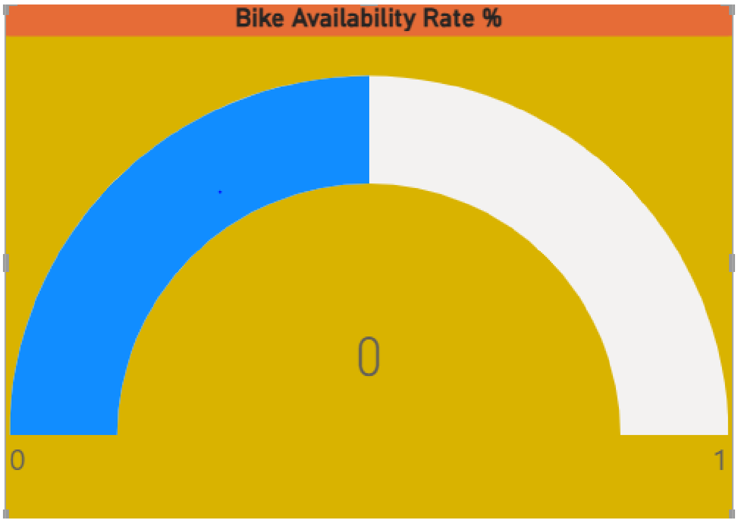


Calculated Dax Measure



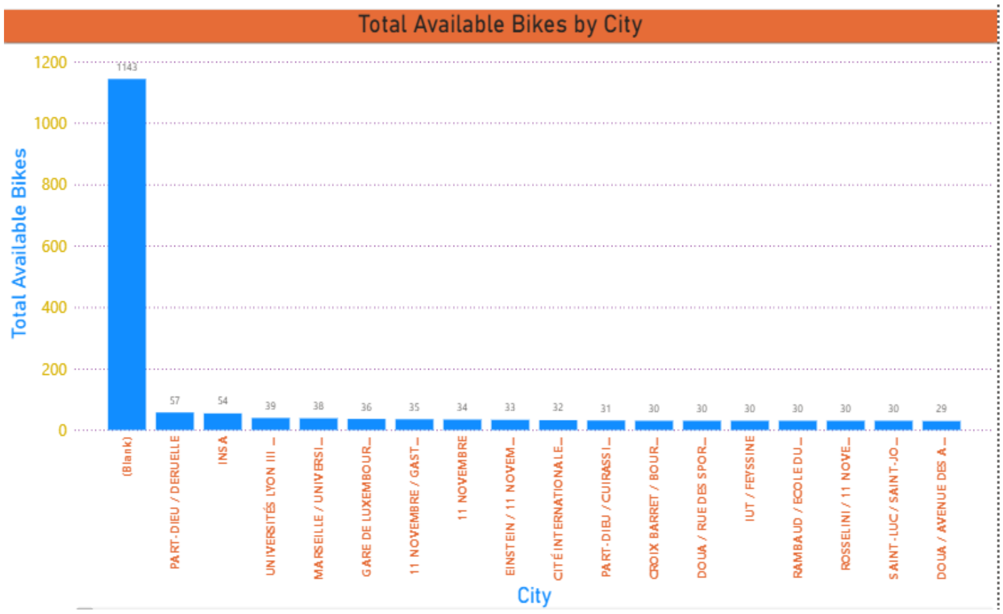
Analaysis and Visualizations(Power BI)

Dashboard Features:



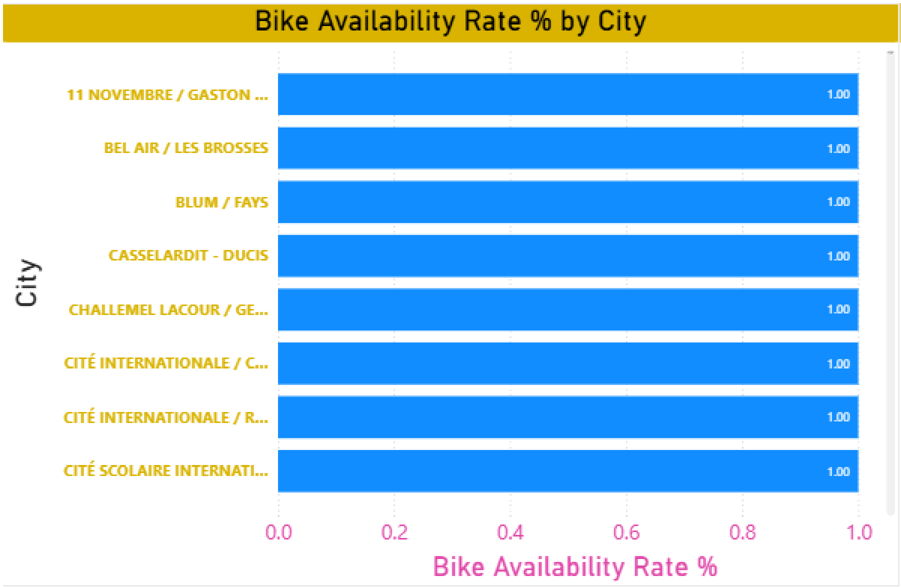
Bike Availability Rate %

- The bike availability rate is critically low, indicating that users are currently unable to find bikes across most stations.
- This near-zero availability highlights a severe imbalance between bike demand and supply in the network.
- Operational intervention is urgently required, such as bike redistribution or fleet expansion, to restore service reliability.
- Sustained low availability at this level can negatively impact user trust and overall system usage.



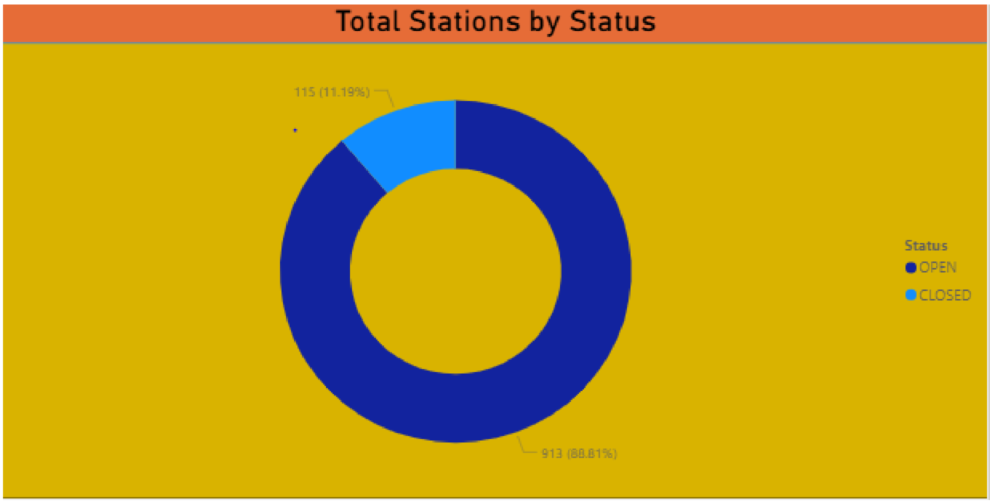
Total Available Bikes by City

- Paris–Île-de-France dominates bike availability, holding an exceptionally higher number of available bikes compared to all other cities.
- All remaining cities show very low and closely clustered bike availability, indicating a strong imbalance in network distribution.
- Cities such as Nantes and Lyon have moderate availability but still fall far behind the leading city, highlighting centralized supply.
- The sharp drop after the top city suggests that bike resources are heavily concentrated rather than evenly allocated across locations.



Bike Availability Rate % by City

- All listed cities show a bike availability rate of 100%, indicating that every bike stand currently has bikes available.
- The uniform availability rate across cities suggests no immediate bike shortages in these locations.
- Such consistently high availability may point to low current demand or recent bike redistribution efforts.
- While availability is optimal, sustained 100% rates could also indicate under-utilization of bikes in these cities.



Total Stations by Status

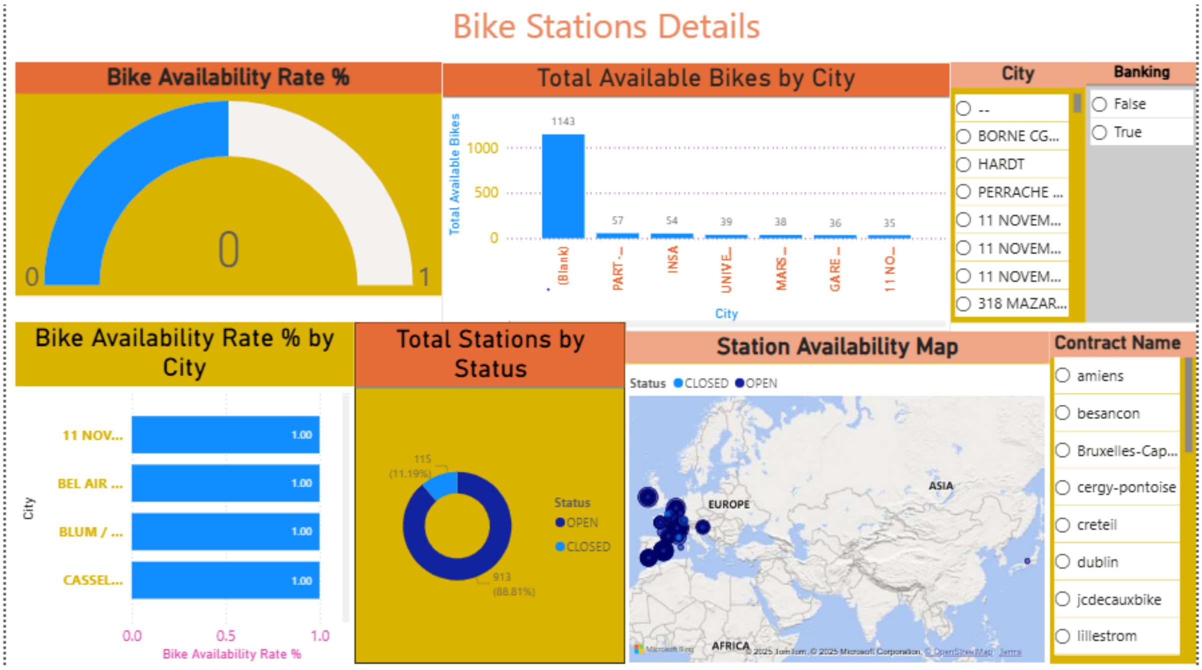
- The vast majority of stations are operational, with approximately 89% marked as OPEN, indicating strong network availability.

- Only about 11% of stations are CLOSED, suggesting limited service disruption across the system.
- The high proportion of active stations reflects effective maintenance and operational management.
- The small share of closed stations highlights targeted opportunities for restoration rather than widespread network issues.

Station Availabililty Map

- Stations are heavily concentrated in Western Europe, indicating this region as the primary operational hub.
- Open stations clearly dominate the map, with closed stations appearing sparse and geographically scattered.
- There is minimal station presence in Asia, Africa, and the Middle East, showing limited geographic coverage beyond Europe.
- High clustering of stations in countries like France, Germany, and nearby regions suggests dense urban or high-demand deployment.

Consolidated Report / Dashboard



Insights & Conclusion

Descriptive Analytics:

- Most bike stations in the network are operational, with OPEN stations significantly outnumbering CLOSED ones.
- Bike availability rates across cities indicate that bikes are generally accessible at most stations.
- Station locations are highly concentrated in European regions, especially in major urban areas.
- Only a small fraction of stations are inactive, reflecting stable operational performance of the bike-sharing system.

Diagnostic Analytics:

- High bike availability rates in several cities indicate lower demand or recent bike redistribution rather than increased supply.
- The small percentage of closed stations suggests that operational downtime is likely due to localized maintenance issues rather than system-wide failures.
- The heavy concentration of stations in certain regions explains the imbalance in bike availability across cities.
- Under-utilization at stations with consistently high availability points to mismatched station placement or timing of demand.

Predictive Analytics:

- Stations with consistently low bike availability are likely to face shortages during peak hours in the near future.
- Cities showing declining availability trends may require additional bikes or station expansion to meet upcoming demand.
- Inactive or low-utilization stations are likely to remain underperforming unless relocated or demand patterns change.
- Seasonal and daily usage patterns can predict periods of high congestion, enabling proactive bike rebalancing.

Prescriptive Analytics:

- Increase bike supply at stations with low availability rates to prevent service disruptions during peak demand.
- Redistribute bikes from low-utilization stations to high-demand areas to optimize overall system efficiency.
- Schedule maintenance or temporary closures for consistently inactive stations to reduce operational costs
- Use demand forecasts to plan dynamic rebalancing routes and staffing for improved station performance.

Conclusion

- This analysis provides clear visibility into bike station availability, utilization, and operational status, helping identify high-demand areas and underperforming stations. The insights support data-driven decisions for bike redistribution, capacity planning, and service optimization, ultimately improving system efficiency and user satisfaction.